

Activation-Patching Effects in Controlled CNNs Depend on Intervention Size, Network Architecture, and First-Layer Filter Structure

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Paper under double-blind review

Abstract

Activation patching measures how a model output changes when internal activations are replaced, but the measured effect depends on how the intervention is defined. We study controlled convolutional networks whose first layer is initialized from a predefined bank of edge, corner, and ring filters. Constant regularization keeps these filters close to the bank, whereas annealing the penalty releases them during training. A pre-specified 20-block test shows that the release-versus-retention difference changes sharply with intervention size, and a separate 100-block study across 16 CNN architectures shows large architecture dependence. A further 100-block pixel-permuted control isolates the two-dimensional arrangement of the predefined filters while preserving the bank’s Gram matrix, rank, singular spectrum, row norms, and coefficient multisets. We then perform a post-hoc decomposition of the saved checkpoints into the sum of singleton channel effects and a residual non-additive term. In analytically additive tiny networks, the observed and additive-reconstruction intervention-size contrasts agree to numerical precision even when the contrast is large; for example, on `two_concepts/tiny_gmp`, probability-error reduction gives $B = 0.832$ in both cases. Thus dependence on intervention size does not by itself imply interactions between patched channels. In deeper networks, especially some BatchNorm variants, observed and additive contrasts can diverge strongly or even change sign. Meanwhile, the structured-versus-pixel-permuted difference is largely preserved by the additive reconstruction: on `two_concepts`, the centered-logit difference averaged across four architectures is $+0.05743$ observed versus $+0.05710$ additive. The results separate three contributors to patching curves: singleton-effect geometry and cancellation, nonlinear evaluation of their sum, and genuine downstream non-additivity. Filter similarity alone therefore does not determine the patching effect, and intervention-size effects should not be interpreted as channel interactions without an explicit additivity check.¹

1 Introduction

The first convolutional layer is one of the few places in a vision network where learned weights can still be inspected directly as spatial filters. This has motivated a long line of work asking whether early spatial filters should be learned freely at all. Wavelet and Gabor constructions provide useful fixed image representations (Bruna & Mallat, 2013). Gabor initialization, maintaining Gabor filter structure during training, learnable Gabor filters, and predefined-filter networks introduce stronger first-layer inductive biases (Özbulak & Ekenel, 2018; Molaei & Shiri Ahmad Abadi, 2020; Wang & Alkhalifah, 2024; Linse et al., 2023; Orozco-Solis et al., 2024). Fixed or even random spatial kernels can also remain effective when later layers learn how to recombine them (Gaudio et al., 2023; Gavrikov & Keuper, 2023). Analyses of unconstrained CNNs likewise find regular first-layer frequency structure and sensitivity of early filters to training-data statistics (Chowers & Weiss,

¹ChatGPT (OpenAI) was used as an assistive tool during experimental planning, code development and debugging, manuscript organization, and language editing. All experimental protocols, analyses, interpretations, and manuscript text were reviewed by the authors, who take full responsibility for the work.

2023; Jorgenson et al., 2026). These results motivate structured first-layer filters, but they leave open a different question: if the first layer remains close to the predefined filters, does that similarity correspond to a consistent patching effect?

The distinction matters because representational alignment and effects on model behavior are not interchangeable. Network Dissection, TCAV, Concept Whitening, concept bottlenecks, and related approaches measure or impose different forms of alignment between internal representations and human-defined concepts (Bau et al., 2017; Kim et al., 2018; Chen et al., 2020; Koh et al., 2020). Subsequent work shows that apparently aligned representations can leak unintended information, fail to correspond to intended semantics, or dissociate from effects on model outputs (Margeloiu et al., 2021; Havasi et al., 2022; Marconato et al., 2022; Nicolson et al., 2025; Sharma & Le, 2026). A filter can therefore remain similar to a predefined spatial filter without implying that interventions on the corresponding activation channels behave in a simple or stable way.

Activation patching provides such an intervention: replace an internal activation from one run with the corresponding activation from another and measure the downstream output change. Yet the measured patching effect is not invariant to the experimental setup. Its interpretation depends on patch direction and scope, component granularity, source and target inputs, and evaluation metric (Heimersheim & Nanda, 2024; Zhang & Nanda, 2024). More generally, circuit-faithfulness scores can change under reasonable changes in ablation methodology, so the evaluation procedure partly defines the question being asked (Miller et al., 2024). These concerns are well established in mechanistic-interpretability work, but much less is known about how they interact with explicitly structured first-layer filters in a controlled vision setting.

We use small CNNs and procedural image tasks to isolate this question. Every model begins with the same 16-channel, 9×9 first convolution. One condition is kept near a predefined bank of edge, corner, and ring filters by a constant penalty; another begins at the same bank but is released by annealing that penalty to zero. Matched counterfactual image pairs let us replace complete first-layer feature maps using activations from another valid input rather than an artificially corrupted image. We ask four questions. Does the difference in patching effect between release and retention depend on intervention size? With the patching location fixed, does changing the network architecture after the first layer change that difference? Is any such effect specific to the original two-dimensional arrangement of the predefined filters, rather than to the Gram matrix, row norms, rank, singular spectrum, or coefficient multisets preserved by the control? Finally, how much of the intervention-size dependence can be reconstructed from singleton channel effects, and how much requires a genuinely non-additive downstream response to joint patching?

We do not interpret structured filters as inherently interpretable. Instead, we test when preserving their structure changes the measured patching effect. Our main findings are:

1. A pre-specified test on independent renderer blocks shows that the difference in patching effect between release and retention depends strongly on intervention size. The pre-specified contrast between $k = 1, 2$ and $k = 4, 8$ remains positive under centered-logit fidelity and unnormalized probability error reduction, although the detailed pointwise curves depend on the metric.
2. A separate pre-specified study across 16 CNN architectures shows large differences across architectures under both evaluation metrics. The same conclusion holds after Greenhouse–Geisser correction, within-block permutation tests, Friedman tests, and random channel subsets.
3. A pixel-permuted control experiment on independent renderer blocks isolates first-layer spatial filter structure. Relative to a common pixel permutation that preserves the original bank’s complete Gram matrix, rank, singular spectrum, row norms, and coefficient multisets, the centered-logit difference for channels selected on the validation set lies outside the pre-specified equivalence margin. That effect is architecture-dependent, reverses sign in one architecture, and is substantially smaller for random channel sets.
4. A post-hoc decomposition of the saved checkpoints shows that intervention-size dependence is not synonymous with channel interaction. Exact-additivity controls retain large B values, while deeper architectures introduce architecture-specific residual non-additivity. The structured-versus-pixel-

permuted difference is itself reproduced closely by the additive reconstruction, indicating that much of the spatial-filter effect is already present in the geometry of the singleton channel contributions.

Together, these results separate filter similarity from patching behavior and separate intervention-size dependence from downstream non-additivity.

2 Related work

Structured and predefined spatial filters. Hand-designed spatial filters predate modern CNNs, and fixed wavelet scattering shows that substantial invariance and discriminative structure can be obtained without learning arbitrary spatial kernels (Bruna & Mallat, 2013). More recent CNN work has used Gabor banks as initialization (Özbulak & Ekenel, 2018), explicitly maintained Gabor structure during optimization (Molaei & Shiri Ahmad Abadi, 2020), constrained Gabor parameters in task-specific first layers (Wang & Alkhalifah, 2024), and learned texture branches built from Gabor filters (Zhu et al., 2023). Orozco-Solis et al. show that learnable Gabor filters can lose their initial structure over training and use similarity measures to track that degeneration (Orozco-Solis et al., 2024). At a broader architectural level, networks with predefined or fixed spatial filters can retain substantial predictive performance (Linse et al., 2023; Gaudio et al., 2023), while random frozen convolutions can be effective when learned linear combinations are sufficiently expressive (Gavrikov & Keuper, 2023). Analyses of ordinary CNNs further suggest that first-layer energy profiles are highly regular and approximately whitening (Chowers & Weiss, 2023), and that early-filter updates reflect properties of the training distribution (Jorgenson et al., 2026). This literature establishes that first-layer spatial structure can be useful and persistent. Our question is whether preserving one such structure yields a consistent patching effect.

Alignment, concepts, and effects on model behavior. Concept-based interpretability methods quantify or impose semantic organization at different representational levels. Network Dissection measures unit-level alignment to visual concepts (Bau et al., 2017); TCAV tests directional sensitivity to user-defined concept vectors (Kim et al., 2018); Concept Whitening rotates latent axes toward specified concepts (Chen et al., 2020); and concept bottleneck models expose an intermediate concept representation that can be edited at test time (Koh et al., 2020). These approaches also expose a recurring limitation: an apparently meaningful representation need not be pure, complete, or used in the intended way. Concept bottlenecks can fail semantic alignment or leak task information (Margeloiu et al., 2021; Havasi et al., 2022; Marconato et al., 2022); concept-vector explanations can vary across layers, become entangled, or depend on spatial position (Nicolson et al., 2025); and recent causal analyses distinguish information that is decodable from information that changes model behavior (Sharma & Le, 2026). We adopt the same distinction at a lower level: filter-template similarity is a structural property, not evidence by itself that interventions on the corresponding activation channels have a stable effect on model behavior.

Activation patching and sensitivity to methodological choices. Direct interventions on visual-network units predate current activation-patching practice: Bau et al. activate or suppress internal CNN/GAN units to test their effect on classification and generated visual content (Bau et al., 2020). Activation patching, interchange interventions, and related causal-abstraction tools generalize the idea of intervening on internal states rather than merely decoding them (Geiger et al., 2025). Practical guidance emphasizes that patching direction, component granularity, input choice, and evaluation metric change what evidence an experiment provides (Heimersheim & Nanda, 2024). Zhang and Nanda show that alternative corruption methods and metrics can yield disparate localization results and caution in particular about non-linear probability metrics and out-of-distribution corruptions (Zhang & Nanda, 2024). Miller et al. similarly show that circuit faithfulness depends strongly on ablation methodology and argue that the task assigned to a circuit cannot be separated from the evaluation procedure (Miller et al., 2024). Broader work questions whether mechanistic explanations are unique or identifiable without additional assumptions (Méloux et al., 2025; Sutter et al., 2025). Our contribution is therefore not the general observation that patching is sensitive to methodological choices. Prior work on structured filters primarily studies filter structure or predictive performance, while prior work on activation patching primarily studies intervention methodology. Here we patch the same first-layer activations, vary how long the predefined filter structure is maintained,

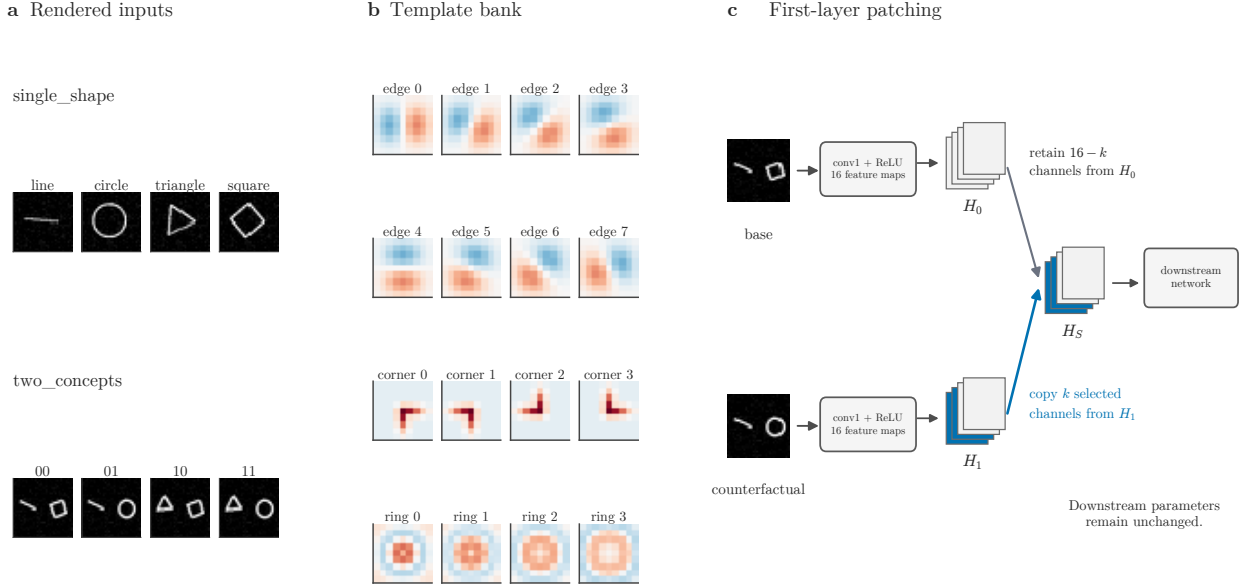


Figure 1: Controlled experimental system. (a) Representative inputs from the procedural renderer. (b) The 16 predefined first-layer spatial filters: eight edges, four corners, and four rings. (c) Activation patching replaces a selected set of k post-ReLU first-layer feature maps in a base run with the corresponding maps from a matched counterfactual run, while downstream parameters remain unchanged. Source activations come from in-distribution counterfactual images; the resulting hybrid activation state is nevertheless an intervention and need not itself lie on the model’s training manifold.

measure how the result changes with intervention size and network architecture, and compare the original two-dimensional arrangement with a control matched on Gram matrix, rank, and singular spectrum. We then add an explicit singleton-additivity baseline so that dependence on intervention size is not conflated with interaction among patched channels.

3 Experimental setup

3.1 Controlled tasks and predefined first-layer filters

We use two 32×32 grayscale procedural classification tasks with matched counterfactual pairs. Each renderer block contains 512 training, 256 validation, and 256 test images. Rendering varies angle, line width, scale, translation, Gaussian noise, and occasional occlusion. In *single_shape*, the four classes are line, circle, triangle, and square. In *two_concepts*, circle and triangle presence define two binary factors and four classes; a counterfactual pair changes one factor while holding the other factor and nuisance variables fixed. Earlier experiments showed a clearer dependence on intervention size on *two_concepts*; subsequent follow-ups on independent renderer blocks therefore designated *two_concepts* as the primary task and retained *single_shape* as a pre-specified secondary task. This task choice was based on the earlier results but specified before evaluation of each later follow-up.

Every model begins with a bias-free $1 \rightarrow 16$ convolution with 9×9 kernels, same padding, and ReLU. The predefined bank contains eight oriented edges, four two-ray corners, and four rings. Each template is centered and unit norm; the 16×81 flattened bank has numerical rank 10. Figure 1 shows the renderer, the bank, and the intervention point.

For learned filter W_i and template T_j , structural similarity is

$$A = \frac{1}{16} \sum_i \max_j \langle \widehat{W}_i, \widehat{T}_j \rangle, \quad (1)$$

where the inner product is signed cosine similarity after centering and normalizing each filter. A one-to-one matching analysis is retained as a secondary structural check.

3.2 Training conditions and filter-bank controls

Template-based models initialize the first convolution from the predefined filter bank W_{ref} and optimize

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda(e) \frac{\|W - W_{\text{ref}}\|_F^2}{\|W_{\text{ref}}\|_F^2}. \quad (2)$$

The central training comparison uses the same initialization but different regularization schedules. *Retention* keeps $\lambda(e) = 1$ for all 200 epochs. *Release* keeps $\lambda = 1$ through epoch 10, decays linearly to zero by epoch 80, and remains zero thereafter. Adam uses learning rate 0.003 and batch size 128, with no learning-rate scheduler or early stopping. This is a contrast between two complete training regimes, not a post-training intervention that changes only the first-layer kernels while holding the remainder of the trained network fixed: downstream parameters co-adapt to the evolving first layer in each regime.

The pixel-permuted control experiment on independent renderer blocks replaces the original reference filter bank with three alternatives while preserving the same retention/release schedules. The primary control, *pixel-permuted*, applies one common random permutation of the 81 spatial coordinates to every row of the structured bank. Because the same permutation is applied to all rows, it preserves the complete 16×16 Gram matrix, row norms, numerical rank, singular spectrum, and each row’s coefficient multiset while destroying the original two-dimensional edge/corner/ring arrangement. Two secondary controls use a generic random rank-10 bank and a generic random full-rank bank. Random filter-bank realizations are paired across task, architecture, and treatment within each block/initialization replicate.

Network architecture variants. The architecture study keeps the first convolution and patching location fixed and varies only the network after the patched layer. Its 16 variants use global max or global average pooling and controlled convolutional depths of 1, 2, or 4. Additional contrasts vary downstream width (16 or 64 channels for plain networks), residual versus plain depth-4 connectivity, and the presence or absence of downstream BatchNorm. All BatchNorm layers occur after the patched first-layer tensor. The pixel-permuted control study uses a balanced four-architecture subset containing the original depth contrast and the pooling factor that showed a clear difference in the completed architecture study: `tiny_gmp`, `tiny_gap`, `plain2_w16_gmp`, and `plain2_w16_gap`. That four-architecture subset was chosen based on the earlier architecture results but specified before any pixel-permuted control outcome was inspected. Exact definitions and parameter counts are reported in Section A.4.

3.3 Activation patching, ranking, and evaluation metrics

For each matched base/counterfactual pair, we cache the post-ReLU first-layer tensor for both images. A patched run starts from the base image and replaces selected feature maps with the corresponding maps from the counterfactual run. No test-set quantity is used to select channels.

Channels are ranked by the absolute standardized validation-set difference in global-max first-layer activation,

$$d_{g,j} = \frac{\bar{z}_{g=1,j} - \bar{z}_{g=0,j}}{s_j + 10^{-6}}, \quad (3)$$

where z_j is the global maximum of feature map j , s_j is its validation-set standard deviation, and g is the factor changed by the counterfactual. For `two_concepts`, g is the binary circle or triangle factor that is toggled in that counterfactual; for `single_shape`, it is the destination shape identity. Channels are ranked by $|d_{g,j}|$ separately for that concept/identity, and the ranking is fixed using validation data before test patching. The

same global-max ranking rule is used for every architecture, including GAP variants, so ranking does not change together with architecture. Eight random channel orders of the same size are evaluated as a control for the validation-based ranking.

Let

$$E_{\text{prob}}(a, b) = \frac{1}{N} \sum_{i=1}^N \|p_a^{(i)} - p_b^{(i)}\|_2^2 \quad (4)$$

denote mean squared probability-space error over N matched pairs. The original patching score is

$$F_{\text{prob}}(S) = 1 - \frac{E_{\text{prob}}(S, 1)}{E_{\text{prob}}(0, 1)}. \quad (5)$$

This score measures normalized reconstruction of the counterfactual output; it is not a mediated fraction or a general measure of causal importance.

Because the denominator in equation (5) varies across models and treatments, later analyses use two alternative evaluation metrics. Centered logits remove the arbitrary common-logit offset,

$$\tilde{\ell} = \ell - \text{mean}(\ell)\mathbf{1}, \quad (6)$$

and centered-logit fidelity applies the same normalized reconstruction form in that space. The unnormalized probability error reduction is

$$R_{\text{prob}}(S) = E_{\text{prob}}(0, 1) - E_{\text{prob}}(S, 1). \quad (7)$$

3.4 Pre-specified contrast and study design

For any patching metric M , define the within-block release-minus-retention difference at intervention size k ,

$$\Delta_b^M(k) = M_b^{\text{release}}(k) - M_b^{\text{retention}}(k), \quad (8)$$

and the pre-specified contrast between $k = 1, 2$ and $k = 4, 8$

$$B_b^M = \frac{\Delta_b^M(4) + \Delta_b^M(8)}{2} - \frac{\Delta_b^M(1) + \Delta_b^M(2)}{2}. \quad (9)$$

A positive B means only that the release-minus-retention difference is larger on average at $k = 4, 8$ than at $k = 1, 2$. It does not imply monotonicity in k or a positive release effect at either larger intervention size.

Matched image pairs are repeated measurements within a model/block and are never treated as independent replicates. The studies were conducted sequentially rather than specified as one joint preregistration. Earlier analyses motivated a pre-specified 20-block **two_concepts** test of the effect of intervention size in TinyCNN. Its architecture contrast and the pre-specified alternative-metric reanalysis then motivated a separate 100-block architecture study with 16 architecture variants. The architecture results then motivated the pixel-permuted control and the balanced four-architecture subset used on a further 100 independent renderer blocks. Each follow-up protocol and its primary analyses were specified before data from its new renderer blocks were evaluated, although the design of each follow-up was informed by the preceding results.

The 16-architecture and pixel-permuted control studies each average four paired initialization or initialization/filter-bank replicates within renderer block before inference. Their inferential sample size is therefore $n = 100$ renderer blocks, not the 25,600 trained models in each grid. The alternative-metric reanalysis of saved checkpoints and the later Greenhouse–Geisser, permutation, Friedman, random-channel, and singleton-additivity analyses are post-hoc with respect to the original outcomes, although their analysis definitions were specified before recomputation. The additivity analysis reuses the completed checkpoints and renderer blocks and is interpreted as mechanistic diagnosis rather than independent confirmation. Earlier and secondary schedule analyses are retained in the appendix but are not pooled with tests on independent renderer blocks.

3.5 Post-hoc additive decomposition of patching effects

After completion of the architecture and pixel-permuted studies, we re-evaluated their saved checkpoints without retraining to ask whether the intervention-size effect requires interactions among patched channels. For a matched pair, let ℓ_0 and ℓ_1 denote the base and counterfactual logits, $\ell_{\{j\}}$ the logits after patching channel j alone, and ℓ_S the logits after jointly patching a channel set S . Define the singleton logit change

$$d_j = \ell_{\{j\}} - \ell_0, \quad (10)$$

the additive reconstruction

$$\widehat{\ell}_S = \ell_0 + \sum_{j \in S} d_j, \quad (11)$$

and the residual

$$r_S = \ell_S - \widehat{\ell}_S. \quad (12)$$

We then apply the same centered-logit and probability-error metrics to both the observed logits ℓ_S and reconstructed logits $\widehat{\ell}_S$. This is important because a nonlinear score can produce a curved intervention-size profile even when $r_S = 0$.

To quantify cancellation among singleton effects, we also report

$$K(S) = 1 - \frac{\left\| \sum_{j \in S} d_j \right\|_2}{\sum_{j \in S} \|d_j\|_2}, \quad (13)$$

whenever the denominator is nonzero. $K(S) = 0$ means the singleton logit changes align perfectly in norm, while larger values indicate partial cancellation. The two tiny architectures provide an analytic implementation control: after the patched first-layer ReLU, their tail is pooling followed by a linear classifier, so whole-channel patching is additive in logits up to numerical error. The decomposition uses the same test pairs, channel rankings, and renderer blocks as the completed source studies. It is therefore a post-hoc mechanistic diagnosis of those outcomes, not an independent confirmatory sample.

4 Results

4.1 Filter similarity does not determine the patching effect

Retention maintains filter similarity as intended. In the larger schedule/task sensitivity sample, **two_concepts**/TinyCNN models under constant regularization have mean template similarity 0.944 versus 0.574 under release, while mean test accuracy is 98.05% versus 99.45%. In the corresponding two-layer model, similarities are 0.999 and 0.896 while accuracies are 99.11% and 99.26%. One-to-one matching gives the same conclusion about filter similarity and rules out the trivial explanation that many learned filters merely match the same few templates (Section E).

Figure 2 provides a representative visualization. It is treated as evidence about filter similarity only: neither visual resemblance nor the score in equation (1) establishes semantic purity or an effect on model behavior.

That difference in filter similarity does not determine the activation-patching effect. In the pre-specified 20-block TinyCNN test, the original probability-fidelity contrast is

$$\bar{B}_{\text{prob}} = 0.33098, \quad 95\% \text{ CI} = [0.27368, 0.38827], \quad (14)$$

with $t_{19} = 12.09$, $p = 2.28 \times 10^{-10}$, and all 20 block-level values positive. The pointwise release-minus-retention differences are -0.3459 , -0.3317 , -0.0221 , and $+0.0065$ at $k = 1, 2, 4, 8$, respectively. Thus a single choice of k would describe only one part of the release-minus-retention comparison.

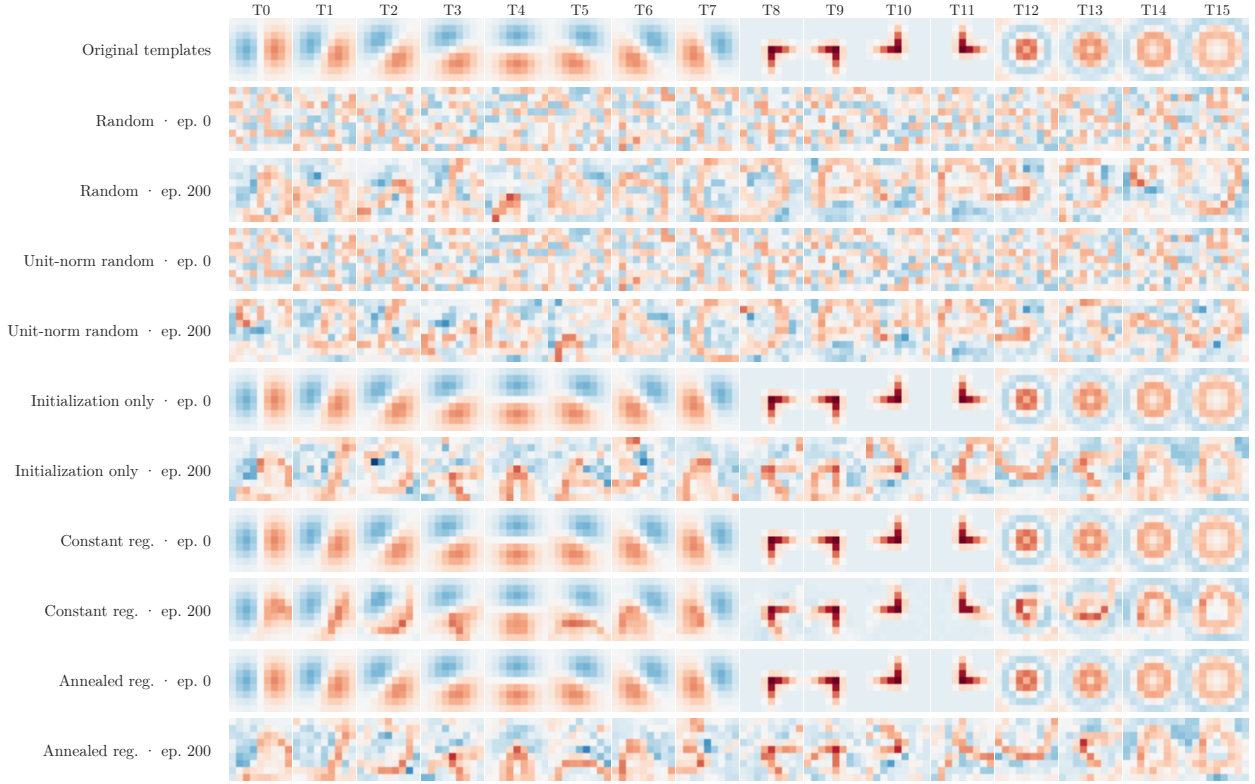


Figure 2: Representative first-layer kernels from a `two_concepts`/TinyCNN checkpoint. The top row shows the predefined bank; later rows show saved filters under the earlier training conditions, reordered by one-to-one template matching for display. Retention preserves visibly template-like kernels, whereas released filters move farther from the bank. The reported statistical results use replicate-level filter-similarity and patching measurements rather than this example.

4.2 Intervention-size contrast across evaluation metrics

The original probability fidelity has a model-specific denominator, and the pre-specified metric reanalysis shows that this denominator changes substantially between retention and release. The denominator $E_{\text{prob}}(0, 1)$ differs between release and retention by $+0.8486$ on average in TinyCNN, while the corresponding centered-logit denominator differs by $+126.54$. The normalization therefore affects the comparison.

Nevertheless, the pre-specified contrast in equation (9) remains positive under both alternative metrics. Centered-logit fidelity gives $B = +0.17888$ (95% CI $[+0.12727, +0.23049]$, Holm $p = 6.95 \times 10^{-7}$), while probability error reduction gives $B = +0.89441$ (95% CI $[+0.78630, +1.00252]$, Holm $p = 8.59 \times 10^{-13}$).

Panel (a) of figure 3 shows that the alternative metrics support the same *sign of the pre-specified B contrast*, but not the same values at each k . In particular, unnormalized probability error reduction becomes strongly positive at $k = 4, 8$, whereas the original normalized probability score is close to zero there. The alternative metrics therefore support the same pre-specified contrast, but they are not interchangeable.

4.3 Activation-patching results depend on network architecture

With the first-layer patching procedure, training conditions, validation ranking, counterfactual pairs, and evaluation metrics fixed, changing only the architecture after the first layer produces large differences:

$$F_{15,1485} = 77.09, \quad \eta_p^2 = 0.438 \quad (15)$$

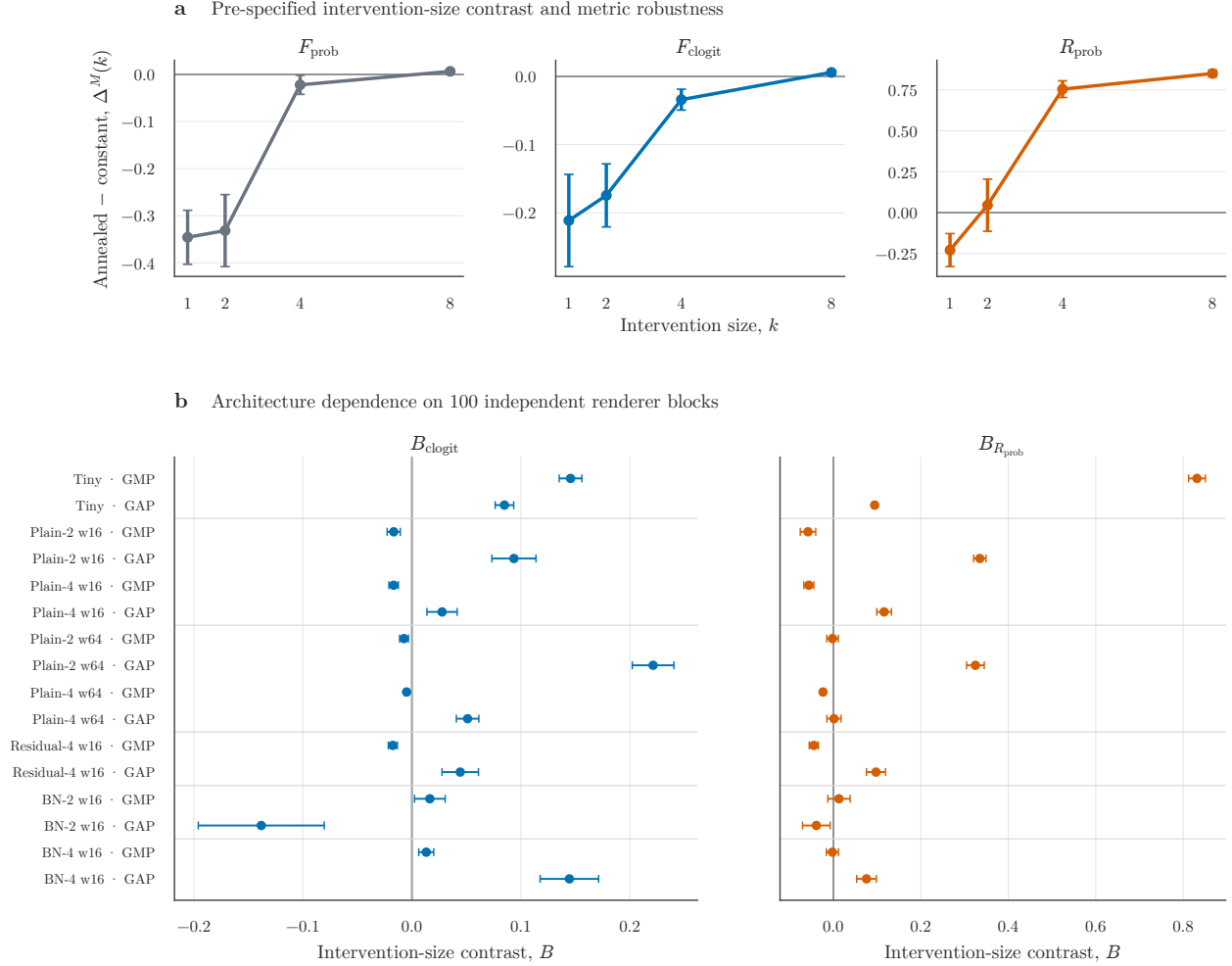


Figure 3: Intervention size and network architecture. **(a)** Release-minus-retention differences $\Delta^M(k)$ in the pre-specified `two_concepts`/TinyCNN experiment under the original normalized probability score, centered-logit fidelity, and unnormalized probability error reduction. The three metrics agree on the sign of B but not on the values at individual intervention sizes. **(b)** Architecture-specific B on the primary `two_concepts` task for 16 pre-specified CNN architectures. Each point averages four initialization replicates within each of 100 independent renderer blocks; bars are marginal 95% t intervals. The two evaluation metrics are shown separately because their numerical scales are not comparable.

for centered-logit fidelity and

$$F_{15,1485} = 656.70, \quad \eta_p^2 = 0.869 \quad (16)$$

for probability error reduction. On these independent blocks, the TinyGMP-minus-Plain2-GMP contrast is +0.16226 (95% CI [+0.14970, +0.17481]) for centered logits and +0.88960 (95% CI [+0.86212, +0.91707]) for probability error reduction.

The pattern in panel (b) of figure 3 is not a single monotone effect of depth or pooling. Under GMP, most of the Tiny-to-plain shift occurs when the first downstream convolution is added; adding further plain depth changes little. Under GAP, the depth pattern differs. GAP-versus-GMP contrasts are large but reverse direction across backbones. Residual-versus-plain depth-4 contrasts are comparatively small, while BatchNorm effects depend on depth, pooling, and metric. These results show architecture dependence rather than a universal effect of any single architectural component.

Two post-hoc analyses, specified before recomputation, test whether the architecture result depends on sphericity assumptions or on the validation-based channel ranking. First, the architecture effect remains significant after Greenhouse–Geisser correction, 100,000 within-block permutations, and Friedman rank tests; the Holm-adjusted permutation p value is 2.0×10^{-5} for both evaluation metrics. Second, differences across architectures remain when B is computed from the mean over eight random channel orders, again with Holm permutation $p = 2.0 \times 10^{-5}$ for both metrics. The architecture result is therefore not solely produced by the validation-derived channel ranking. Ranking still matters quantitatively: Spearman correlation across architecture means for selected versus random channels is 0.256 for centered logits and 0.724 for probability error reduction (Section A.4).

4.4 Effect of the two-dimensional filter arrangement

We next tested whether the effect depended on the original edge/corner/ring arrangement or arose from regularizing the first layer toward a fixed filter bank more generally. The pixel-permuted control experiment addresses this directly. Its primary control applies the same random permutation of spatial coordinates to every template, preserving the full Gram matrix, numerical rank, singular spectrum, row norms, and row coefficient multisets while destroying the original two-dimensional arrangement.

Averaged over the four architectures on `two_concepts`, selected-channel centered-logit B differs between the structured and pixel-permuted filter banks by $+0.05743$ (95% CI $[+0.05116, +0.06369]$, Holm sign-flip $p = 2.0 \times 10^{-5}$). Before these outcomes were inspected, the equivalence margin was pre-specified as 20% of the mean absolute B from the earlier architecture study for the same four architectures. The centered-logit 90% interval $[+0.05219, +0.06266]$ lies entirely outside that pre-specified equivalence margin (± 0.01704), so the effect averaged across architectures differs from zero and lies outside the pre-specified equivalence margin.

For probability error reduction, the difference averaged across architectures is $+0.04563$ (95% CI $[+0.03728, +0.05397]$, Holm sign-flip $p = 2.0 \times 10^{-5}$). Its 90% interval $[+0.03864, +0.05261]$ lies inside the wider pre-specified equivalence margin (± 0.06594). Thus the probability-error contrast averaged across architectures differs from zero, but its 90% interval lies within the pre-specified equivalence margin. This equivalence statement depends on the margin defined from the earlier architecture study; it does not imply a negligible effect of spatial arrangement in every architecture.

The effect also varies strongly across architectures. For channels selected on the validation set, the difference between the structured and pixel-permuted filter banks is positive for `tiny_gmp`, `tiny_gap`, and `plain2_w16_gap`, but negative for `plain2_w16_gmp` under both evaluation metrics. The interaction has partial $\eta^2 = 0.481$ for centered logits and 0.614 for probability error reduction, with Holm permutation $p = 2.0 \times 10^{-5}$ for both. Panel (a) of figure 4 shows the sign reversal directly.

For random channel subsets, the difference in B between the structured and pixel-permuted banks, averaged across architectures, is $+0.00571$ for centered logits and $+0.02007$ for probability error reduction, and both effects fall within the pre-specified equivalence margins. For centered logits, the random-channel difference is smaller than the corresponding difference for channels selected on the validation set ($+0.05743$); for probability error reduction, the same comparison is $+0.02007$ versus $+0.04563$. Architecture-dependent random-channel differences nevertheless remain detectable. The effect of the original spatial arrangement is therefore larger on average for channels selected on the validation set than for random subsets under each evaluation metric, although it is not absent for random subsets.

Finally, the secondary filter-bank controls argue against numerical rank as the main explanation. On the primary task, the pixel-permuted and generic random rank-10 filter banks are generally similar, and differences between the random rank-10 and random full-rank filter banks are generally small after the pre-specified family corrections. The original two-dimensional organization is the filter-bank property most clearly isolated by the selected-channel result, but its sign and practical magnitude remain architecture- and evaluation-metric-dependent.

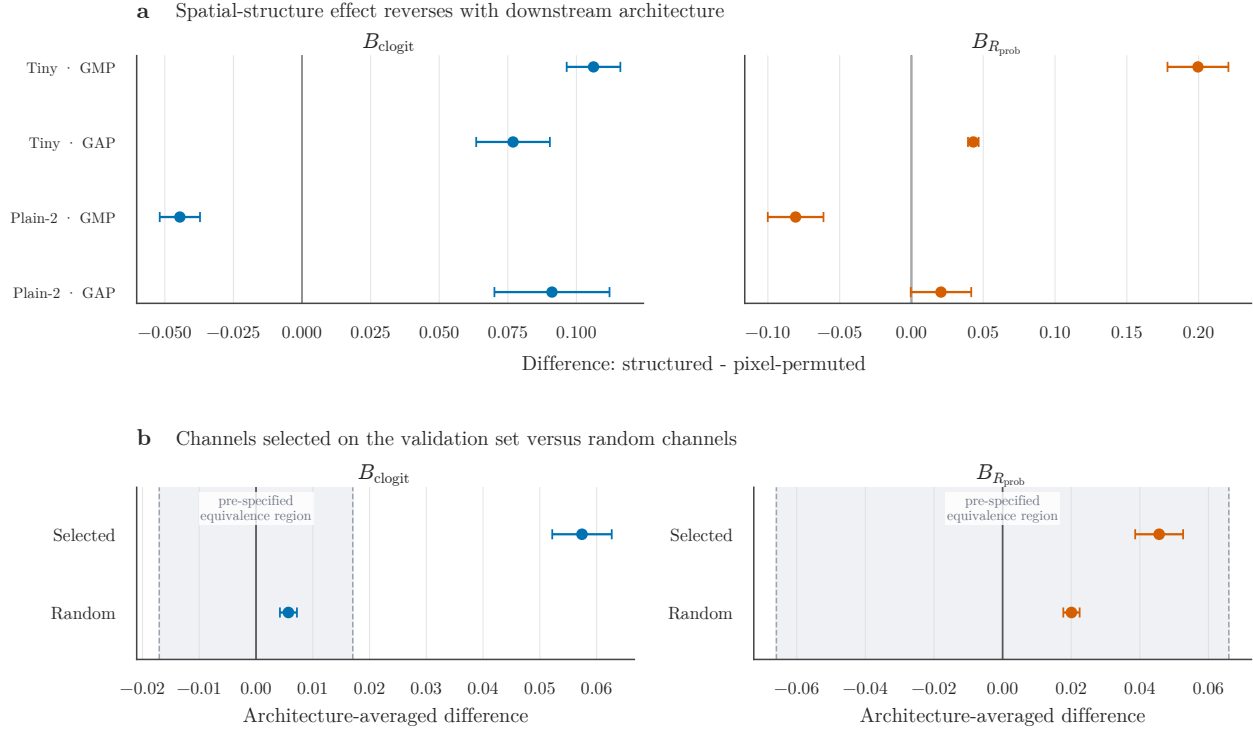


Figure 4: Pixel-permuted control experiment on independent renderer blocks. **(a)** Difference in B between the original structured bank and the pixel-permuted bank for channels selected on the validation set, shown separately for each architecture on `two_concepts`. The spatial effect reverses sign in `plain2_w16_gmp`. Bars are paired 95% t intervals over 100 renderer blocks. **(b)** Differences between the structured and pixel-permuted banks, averaged across architectures, for channels selected on the validation set and for random channel sets. Bars are 90% intervals used by the pre-specified TOST procedure; shaded regions denote the pre-specified equivalence margins, defined as 20% of the mean absolute B from the earlier architecture study. The centered-logit average for channels selected on the validation set is not equivalent, whereas the probability-error average for those channels and both random-channel averages fall within their respective pre-specified margins.

4.5 Intervention-size effects can arise without channel interactions

The post-hoc decomposition covers all 25,600 checkpoints from the architecture study and the 12,800 primary structured/pixel-permuted checkpoints from the spatial-control study. Full-patch and no-op logit identities are exact in the archived analysis, and the analytic tiny-network control behaves as expected.

The first result is a caution about interpretation: a large intervention-size contrast does not require downstream channel interactions. On `two_concepts/tiny_gmp` in the architecture study, centered-logit fidelity gives $B = +0.14558$ for both the observed and additive-reconstructed curves, while probability error reduction gives $B = +0.83151$ for both. The same observed-versus-additive equality holds throughout the tiny architectures to numerical precision. Yet their intervention-size profiles are far from flat. The curvature therefore arises from the geometry of the singleton logit changes and from evaluating their sum with a nonlinear score, not from a nonzero residual r_S .

The singleton effects also increasingly cancel as more channels are patched. For example, in the structured-filter, release condition on `single_shape`, the mean cancellation score across the four spatial-control architectures rises from about 0.20 at $k = 4$ to 0.32 at $k = 8$ and 0.45 at $k = 16$. Thus the contribution of a larger channel set cannot generally be understood by adding singleton magnitudes: the directions of the singleton logit changes matter.

Deeper networks do introduce genuine non-additivity, and its magnitude is architecture-dependent. The clearest cases are some BatchNorm variants. On `single_shape/bn2_w16_gap`, centered-logit B is -0.27012 for the observed curve but $+0.34955$ under the additive reconstruction, reversing sign. On `bn4_w16_gap`, the corresponding values are $+0.37725$ observed and $+1.64910$ additive. These discrepancies cannot be explained by metric nonlinearity alone; they require a substantial residual in the jointly patched computation. Other plain and residual architectures are generally closer to the additive reconstruction, although not exactly additive.

Finally, the spatial-filter effect is largely already present in the additive component. Averaged over the four spatial-control architectures on `two_concepts`, the structured-minus-pixel-permuted difference in centered-logit B is $+0.05743$ for the observed curves and $+0.05710$ for the additive reconstructions. For probability error reduction the corresponding values are $+0.04563$ and $+0.04105$. The same qualitative agreement holds on `single_shape`. This suggests that preserving the original two-dimensional filter arrangement primarily changes the singleton channel contributions and how they combine, while downstream non-additivity modulates that effect in an architecture-specific way.

4.6 Checks and secondary analyses

The pixel-permuted control experiment on independent renderer blocks completes all 25,600 planned training jobs and evaluations. Maximum full-patch and no-op logit identity errors are zero, and the maximum structured-versus-pixel-permuted Gram-matrix discrepancy is 6.66×10^{-16} . The earlier independent checkpoint audit also reproduces central metrics from the earlier analyses at 32 fixed checkpoints within pre-specified numerical tolerances. Appendix endpoint summaries report test accuracy and similarity to each condition’s own filter bank for the final spatial-control study, together with denominator shifts across the 16-architecture study (Tables 2 and 11). Secondary schedule, task, ranking, and energy-matched analyses are reported in the appendix as supporting analyses and are not treated as independent replications of the tests on independent renderer blocks.

5 Discussion

Maintaining first-layer filter structure, observing an intervention-size effect, and demonstrating interactions among patched channels are three different claims. Work on predefined, fixed, and Gabor-based filters shows that spatial structure can be useful, robust, or even unnecessary for predictive performance depending on how later layers recombine it (Linse et al., 2023; Gaudio et al., 2023; Gavrikov & Keuper, 2023; Molaei & Shiri Ahmad Abadi, 2020). Our retention condition confirms that the predefined structure can be maintained almost exactly, but high filter similarity does not determine a unique patching effect.

The additive decomposition sharpens the interpretation of the intervention-size results. The exact-additivity tiny controls retain large release-minus-retention contrasts even though jointly patching channels is exactly the sum of their singleton logit effects. Intervention-size dependence is therefore not evidence by itself for synergy, redundancy, or another nonlinear interaction among channels. It can arise because singleton effects point in different directions and because fidelity or probability metrics are nonlinear functions of their summed logit change. The growing cancellation scores provide a direct geometric example of this effect.

At the same time, the architecture study is not reducible to metric geometry. In several deeper networks, especially BatchNorm variants, observed and additive-reconstructed contrasts differ substantially and can even have opposite signs. Those cases identify genuine downstream non-additivity in the jointly patched computation. The architecture result can therefore be read more precisely: architecture controls how much of the intervention-size profile is inherited from additive singleton geometry and how much is introduced by nonlinear downstream processing.

The pixel-permuted control adds a complementary result. Under centered-logit fidelity, selected-channel B averaged across architectures differs clearly between the structured and pixel-permuted filter banks, and the new decomposition reproduces almost the same structured-minus-permuted difference using only the summed singleton effects. The original two-dimensional arrangement therefore changes the functional geometry of the first-layer channels before strong downstream interaction is required. This is not a universal benefit of

the original templates: the effect reverses in one architecture, the probability-error contrast averaged across architectures falls within the pre-specified equivalence margin, and the random-channel contrasts averaged across architectures are smaller.

The retention-versus-release contrast should still not be read as the isolated causal effect of first-layer geometry in an otherwise fixed trained network. The regularization schedule changes optimization throughout training, and downstream weights co-adapt to the evolving first layer. The comparison concerns two complete training regimes and asks how their activation-patching effects differ.

These distinctions parallel a broader lesson from concept-based and mechanistic interpretability. Alignment, decodability, and causal influence answer different questions (Margeloiu et al., 2021; Nicolson et al., 2025; Sharma & Le, 2026). Likewise, activation patching is a family of interventions whose conclusions depend on intervention and evaluation choices (Heimersheim & Nanda, 2024; Zhang & Nanda, 2024; Miller et al., 2024). Our experiments add a practical diagnostic: when conclusions change with intervention size, singleton reconstruction should be checked before interpreting that dependence as evidence for interactions among components.

These results suggest reporting intervention size, the type of patched component, evaluation metric, channel-selection rule, network architecture, and source and target inputs alongside activation-patching results. Filter similarity should be reported separately from patching effects, and observed multi-channel effects should be compared with an additive singleton baseline when mechanistic interaction is part of the claim. When a null comparison is scientifically meaningful, equivalence should be tested rather than inferred from nonsignificance.

6 Limitations

- The experiments use two related synthetic image tasks, one family of first-layer spatial templates, and small CNNs. Matched counterfactual pairs provide tight experimental control, but the results do not establish natural-image, large-CNN, vision-transformer, or language-model generalization.
- Test accuracy varies across architectures in the final spatial-control grid, especially for some GAP variants (Table 2). The architecture comparisons are therefore not performance-matched: they characterize the trained systems in the grid rather than an architecture effect at fixed predictive performance or confidence.
- Patching replaces complete post-ReLU feature maps with activations from an in-distribution matched counterfactual. The source activations are therefore not generated by artificial input corruption, but the hybrid activation state can still be off-manifold for the trained model.
- The validation ranking uses global-max first-layer activation for every architecture, including GAP variants. This keeps ranking fixed while architecture changes, but it may favor some representations. Random-channel analyses show that architecture dependence remains without that ranking, while the pixel-permuted control experiment on independent renderer blocks shows that the effect of spatial arrangement is much smaller for random channels.
- The original normalized probability score has a model-specific denominator that differs across treatments. Later analyses therefore use centered-logit and unnormalized probability evaluation metrics. Agreement is strongest at the level of the pre-specified B contrast and architecture omnibus; values at individual intervention sizes and conclusions about effect size can differ across metrics.
- The probability-error difference between the structured and pixel-permuted banks, averaged across architectures, differs from zero but falls within the pre-specified equivalence margin, defined as 20% of the mean absolute B from the earlier architecture study. That margin was pre-specified before outcomes on the independent renderer blocks were inspected, but it is a study convention tied to the effect scale in the earlier architecture study rather than an externally validated threshold of scientific importance; architecture-specific effects can be larger and can change sign.

- The studies were conducted sequentially. The architecture study was motivated by earlier outcomes, and the pixel-permuted-control task/architecture subset was chosen after inspecting the architecture results. Each follow-up used a protocol specified before evaluation on independent renderer blocks, but the sequence should not be interpreted as one jointly preregistered experiment.
- The additive decomposition is post-hoc and reuses the same checkpoints, test pairs, rankings, and renderer blocks as the architecture and spatial-control studies. It clarifies which parts of the observed curves can be reconstructed from singleton effects, but it is not an independent replication. Moreover, a nonzero output-logit residual identifies failure of singleton superposition without uniquely localizing where in the downstream network that interaction arises.
- Retention versus release compares complete training regimes. Because downstream parameters co-adapt during training, the contrast does not identify the direct causal effect of changing only final first-layer geometry while holding the rest of the trained network fixed.
- Analyses of individual architecture factors and the later Greenhouse–Geisser, Friedman, filter-similarity, and random-channel analyses include tests specified before recomputation, but some are post-hoc with respect to the original architecture outcomes. They are supporting analyses, not independent prospective confirmation.
- Filter-template similarity is a geometric score over filter weights. Neither that score nor the visual appearance of filters establishes human interpretability, semantic purity, a unique mechanism, or an identifiable causal abstraction (Méloux et al., 2025; Sutter et al., 2025).

7 Conclusion

Preserving a predefined first-layer filter bank does not determine the patching effect, and dependence on intervention size does not by itself demonstrate interactions among patched channels. In controlled CNNs with matched counterfactuals, large intervention-size contrasts persist in analytically additive networks because singleton channel effects can cancel and the evaluation metric is nonlinear in their sum. Deeper architectures add a second component: genuine downstream non-additivity, which can be large enough to change the magnitude or sign of the release-versus-retention contrast. A pixel-permuted control further shows that the original two-dimensional edge/corner/ring arrangement changes the patching effect beyond the bank’s Gram matrix, rank, and singular spectrum, and that most of this structured-versus-permuted difference is already present in the additive singleton reconstruction. Activation-patching studies should therefore separate structural alignment, singleton-effect geometry, metric dependence, and residual multi-component interaction rather than treating a multi-channel patching curve as a single mechanistic quantity.

8 Data and code availability

The experiments use procedurally rendered synthetic images. An anonymized supplementary ZIP accompanying this submission contains the renderer, model and filter-bank definitions, training and evaluation code, pre-specified protocols, result tables, analysis scripts, checks for the pixel-permuted control, and the independent checkpoint audit. Large raw checkpoints and per-model evaluator outputs are stored separately, while the aggregate tables for the additive decomposition, reconstruction gaps, renderer-block bootstrap intervals, and all data needed to reproduce the reported statistics and figures are included or versioned in the supplementary artifact. For anonymous review, the manuscript does not link the current working repository because it identifies the authors.

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Appendix organization. Section A reports the pixel-permuted control experiment and the post-hoc architecture analyses. Section B gives the complete metric- and architecture-robustness summaries. Section C reports the pre-specified intervention-size study and the schedule/task sensitivity analyses. Section D preserves the earlier analyses that motivated the pre-specified contrast. Section E reports the complete filter-matching table.

A Pixel-permuted control and architecture analyses

This appendix reports the complete numerical results for the pixel-permuted control experiment and the post-hoc architecture analyses. The control experiment was specified before evaluation on independent renderer blocks; the architecture analyses were performed post-hoc after the original architecture results were known and are labeled accordingly.

A.1 Pixel-permuted control experiment

The experiment uses 100 independent renderer blocks, four initialization/filter-bank replicates per block, two tasks, four architectures, four filter-bank families, and two regularization schedules, for 25,600 trained and evaluated models. The independent unit for inference is the renderer block after averaging the four within-block replicates.

The primary comparison is the original structured edge/corner/ring bank against a bank formed by applying one common random permutation of the 81 spatial coordinates to every template row. This pixel-permuted control preserves, to numerical tolerance, every row norm, the complete 16×16 Gram matrix, numerical rank, singular spectrum, and row coefficient multisets while destroying the original 2D spatial arrangement.

Table 1: Difference in B between the structured and pixel-permuted filter banks for channels selected on the validation set, averaged across architectures on `two_concepts`. Difference tests use 100,000 sign-flip permutations and Holm correction across the two evaluation metrics. Equivalence uses the pre-specified TOST margins and 90% confidence intervals.

Metric	Mean difference	95% CI	Holm diff. p	Equivalence result
Centered-logit fidelity	+0.05743	[0.05116, 0.06369]	2.0×10^{-5}	not equivalent ($\delta = 0.01704$)
Probability error reduction	+0.04563	[0.03728, 0.05397]	2.0×10^{-5}	equivalent ($\delta = 0.06594$)

The equivalence margins were pre-specified before the new outcomes as 20% of the mean absolute B from the earlier architecture study for the same four architectures: $\delta_{\text{clogit}} = 0.01704$ and $\delta_{R_{\text{prob}}} = 0.06594$. These are study-specific equivalence margins rather than externally validated thresholds of scientific importance. For centered logits, the 90% interval averaged across architectures is [0.05219, 0.06266] and the Holm-adjusted TOST p value is 1. For probability error reduction, the 90% interval averaged across architectures is [0.03864, 0.05261] and the Holm-adjusted TOST p value is 5.03×10^{-6} . Thus the probability-error difference averaged across architectures differs from zero while falling inside the pre-specified equivalence region. The smallest symmetric margin that would contain its observed 90% interval is approximately 0.05261, about 16.0% of the effect scale from the earlier architecture study used to define the pre-specified 20% margin; this sensitivity is reported to make the role of the margin explicit.

Model endpoints for the primary control task. To make model competence and filter retention directly inspectable, Table 2 reports mean test accuracy and similarity to each model’s own reference bank for the structured and pixel-permuted conditions on `two_concepts`. Each entry first averages the four initialization/filter-bank replicates within renderer block and then averages over the 100 blocks. The table also makes clear that the architecture grid is not performance-matched: the GAP variants, especially `tiny_gap`, have lower accuracy than the GMP variants.

Table 2: Endpoint audit for the four architectures in the pixel-permuted control on **two_concepts**. Similarity is measured against the corresponding reference bank for that row, not always against the original structured bank.

Architecture	Reference bank	Acc. retention	Acc. release	Similarity retention	Similarity release
tiny_gmp	structured	97.83%	99.43%	0.943	0.573
tiny_gmp	pixel-permuted	97.59%	99.31%	0.936	0.563
tiny_gap	structured	43.26%	79.04%	0.999	0.590
tiny_gap	pixel-permuted	41.17%	83.29%	0.999	0.422
plain2_w16_gmp	structured	99.07%	99.31%	0.999	0.900
plain2_w16_gmp	pixel-permuted	99.21%	99.31%	0.999	0.908
plain2_w16_gap	structured	82.61%	92.97%	0.974	0.731
plain2_w16_gap	pixel-permuted	85.33%	92.96%	0.961	0.668

A.2 Architecture dependence of the effect of spatial arrangement

The difference between the structured and pixel-permuted filter banks varies strongly across the four pre-specified architectures.

Table 3: Difference in B between the structured and pixel-permuted filter banks for channels selected on the validation set, shown separately for each architecture on **two_concepts**. Intervals are paired 95% t intervals over 100 renderer blocks.

Architecture	Centered-logit diff.	95% CI	Probability-error diff.	95% CI
tiny_gmp	+0.10624	[0.09645, 0.11602]	+0.19953	[0.17831, 0.22075]
tiny_gap	+0.07690	[0.06348, 0.09033]	+0.04308	[0.03938, 0.04678]
plain2_w16_gmp	-0.04453	[-0.05187, -0.03719]	-0.08067	[-0.10006, -0.06127]
plain2_w16_gap	+0.09109	[0.07012, 0.11207]	+0.02057	[-0.00051, 0.04164]

The corresponding within-block architecture interaction is $F(3, 297) = 91.79$, partial $\eta^2 = 0.481$, Holm permutation $p = 2.0 \times 10^{-5}$ for centered-logit fidelity, and $F(3, 297) = 157.70$, partial $\eta^2 = 0.614$, Holm permutation $p = 2.0 \times 10^{-5}$ for probability error reduction. The sign reversal in **plain2_w16_gmp** rules out a universal positive effect of the original spatial arrangement.

A.3 Random-channel control

Repeating the primary comparison after averaging over eight random channel orders produces much smaller effects of the original spatial arrangement.

Table 4: Difference in B between the structured and pixel-permuted filter banks for random channel sets, averaged across architectures on **two_concepts**. Both confidence intervals exclude zero and fall within the pre-specified equivalence margins.

Metric	Mean difference	95% CI
Centered-logit fidelity	+0.00571	[0.00392, 0.00750]
Probability error reduction	+0.02007	[0.01722, 0.02291]

The interaction between filter-bank type and architecture for random channels remains detectable but is smaller for centered logits: partial $\eta^2 = 0.0683$ with permutation $p = 8.0 \times 10^{-5}$. For probability error reduction it remains substantial, partial $\eta^2 = 0.2910$ with permutation $p = 1.0 \times 10^{-5}$. Thus small effects for random channel sets, averaged across architectures, coexist with architecture-dependent differences; the equivalence statements apply to the averages across architectures, not to every architecture separately. These are pre-specified secondary analyses.

Secondary filter-bank contrasts show that the pixel-permuted and generic random rank-10 banks are generally similar on the primary task, and that the random rank-10 and random full-rank banks are also generally similar after the pre-specified Holm corrections. The original 2D arrangement, rather than numerical rank alone, is therefore the filter-bank property most clearly distinguished by the selected-channel analysis.

The full-patch and no-op checks give zero logit error, as required. The maximum Gram-matrix discrepancy between the structured and pixel-permuted banks is 6.66×10^{-16} .

A.4 Post-hoc architecture analyses

Exact architecture definitions. All variants share the same patched $1 \rightarrow 16$, 9×9 first convolution and ReLU. Pooling is global max (GMP) or global average (GAP). Plain depth-2 networks add one 3×3 convolution; plain depth-4 networks add three. Width-64 variants expand the first downstream convolution to 64 channels and remain width 64 thereafter. Residual depth-4 variants keep width 16 and add the third downstream convolution to the original first-layer activation before the final ReLU. BatchNorm, when present, is downstream of the intervention tensor.

Table 5: Exact architecture grid used in the pre-specified architecture study. Parameter counts include the classifier.

Architecture	Conv depth	Width	Pool	BN	Connectivity	Parameters
tiny_gmp	1	16	GMP	no	none	1,364
tiny_gap	1	16	GAP	no	none	1,364
plain2_w16_gmp	2	16	GMP	no	plain	3,684
plain2_w16_gap	2	16	GAP	no	plain	3,684
plain4_w16_gmp	4	16	GMP	no	plain	8,324
plain4_w16_gap	4	16	GAP	no	plain	8,324
plain2_w64_gmp	2	64	GMP	no	plain	10,836
plain2_w64_gap	2	64	GAP	no	plain	10,836
plain4_w64_gmp	4	64	GMP	no	plain	84,692
plain4_w64_gap	4	64	GAP	no	plain	84,692
res4_w16_gmp	4	16	GMP	no	residual	8,324
res4_w16_gap	4	16	GAP	no	residual	8,324
bn2_w16_gmp	2	16	GMP	yes	plain	3,700
bn2_w16_gap	2	16	GAP	yes	plain	3,700
bn4_w16_gmp	4	16	GMP	yes	plain	8,372
bn4_w16_gap	4	16	GAP	yes	plain	8,372

The original 16-architecture study was pre-specified. After seeing its results, separate post-hoc analyses tested whether the omnibus conclusion depended on the uncorrected sphericity assumption or on the validation-based channel ranking.

Table 6: Sphericity-corrected and permutation-based architecture omnibus analyses on the primary **two_concepts** task. Permutation p values use 100,000 within-block permutations and are Holm adjusted across the two metrics within each analysis kind.

Metric	GG ϵ	GG p	Permutation	Holm p	Friedman p
Centered-logit fidelity	0.1861	1.56×10^{-34}		2.0×10^{-5}	3.52×10^{-146}
Probability error reduction	0.5973	$< 10^{-300}$		2.0×10^{-5}	7.79×10^{-200}

Differences across architectures also remain when B is computed from the mean over eight random channel orders: Holm-adjusted permutation $p = 2.0 \times 10^{-5}$ for both evaluation metrics. The Spearman correlation across architecture means for selected versus random channels is 0.256 for centered logits and 0.724 for probability error reduction.

For the TinyGMP-minus-Plain2-GMP contrast using random channel sets, the mean difference is +0.002765 (95% CI [+0.000014, +0.005515], Holm $p = 0.0488$) for centered logits and +0.319871 (95% CI [+0.312788, +0.326953], Holm $p = 2.76 \times 10^{-96}$) for probability error reduction. The architecture conclusion is therefore not solely a consequence of the selected-channel ranking, although the ranking changes effect magnitude and the ordering across architectures.

B Metric sensitivity and architecture results

This appendix gives the complete numerical results for the pre-specified metric-sensitivity reanalysis and the pre-specified architecture study. Block-level values, supporting tables, and control analyses are versioned under `analysis/metric_sensitivity_001/` and `analysis/architecture_robustness_001/`. The main text reports the principal estimates.

B.1 Metric-sensitivity analysis

The metric-sensitivity analysis reuses saved checkpoints from the pre-specified intervention-size and schedule/task sensitivity experiments. It was designed after the original probability-fidelity results were known, so it is post-hoc with respect to those results; however, the alternative metric definitions and analysis plan were specified before their values were computed. No model was retrained and the original validation-derived channel rankings were reused.

For the primary `two_concepts`/TinyCNN setting, Table 7 gives the full pointwise annealed-minus-constant curves under the two alternative evaluation metrics.

Table 7: TinyCNN pointwise release-minus-retention contrasts under the pre-specified alternative metrics.

Metric	k	Mean Δ	95% CI
Centered-logit fidelity	1	-0.21134	[-0.27899, -0.14369]
Centered-logit fidelity	2	-0.17452	[-0.22078, -0.12825]
Centered-logit fidelity	4	-0.03405	[-0.04939, -0.01871]
Centered-logit fidelity	8	+0.00595	[+0.00253, +0.00937]
Probability error reduction	1	-0.22942	[-0.33033, -0.12852]
Probability error reduction	2	+0.04493	[-0.11443, +0.20428]
Probability error reduction	4	+0.75429	[+0.70327, +0.80530]
Probability error reduction	8	+0.85004	[+0.82915, +0.87093]

The corresponding B contrasts are +0.178881 (95% CI [+0.127268, +0.230494]) for centered-logit fidelity and +0.894411 (95% CI [+0.786301, +1.002522]) for probability error reduction. Holm-adjusted p values for the two-metric family are 6.95×10^{-7} and 8.59×10^{-13} , respectively.

The base-to-counterfactual error denominator differs between release and retention. In TinyCNN, release minus retention is +0.848558 (95% CI [+0.826701, +0.870416]) for the probability-space denominator and +126.537 (95% CI [+120.845, +132.229]) for the centered-logit denominator. In TwoLayerCNN the corresponding differences are +0.062062 and +211.368. Thus the original normalized metric cannot be treated as though its denominator were constant across training conditions.

The same pre-specified analysis applied to the schedule/task sensitivity experiment gives the default-release results in Table 8.

Table 8: Schedule/task sensitivity default-release B contrasts under the pre-specified alternative metrics. Values are release minus constant $\lambda = 1$ under the original activation-contrast ranking.

Task	Architecture	B_{clogit}	$B_{R_{\text{prob}}}$
single shape	TinyCNN	+0.05660	+0.33747
single shape	TwoLayerCNN	+0.05700	-0.01622
two concepts	TinyCNN	+0.13427	+0.80239
two concepts	TwoLayerCNN	-0.01953	-0.05497

The maximum absolute discrepancy between recomputed and stored original probability fidelity is 1.24×10^{-7} . Maximum full-patch and no-op logit identity errors are zero.

B.2 Architecture results on the primary task

The architecture study uses 100 independent renderer blocks, four paired initialization replicates per block, 16 CNN architectures, two tasks, and the two central training conditions. Initialization replicates are averaged before inference. The primary task is `two_concepts`; the independent unit for inference is therefore the renderer block ($n = 100$) for each architecture-level comparison.

Table 9: Repeated-measures architecture omnibus tests on `two_concepts`.

Metric	F	df	partial η^2	Holm p
centered_logit_fidelity	77.09	15, 1485	0.438	2.40×10^{-173}
prob_error_reduction	656.70	15, 1485	0.869	0 (machine precision)

For probability error reduction, the computed upper-tail p value is numerically zero at double precision; this is reported as a machine-precision underflow rather than a literal probability of zero.

Table 10: TinyGMP-minus-Plain2-GMP architecture contrast, `tiny_gmp` minus `plain2_w16_gmp`.

Metric	Mean difference in B	95% CI	Holm p
centered_logit_fidelity	+0.16226	[+0.14970, +0.17481]	1.66×10^{-45}
prob_error_reduction	+0.88960	[+0.86212, +0.91707]	3.24×10^{-82}

Input-effect denominators across architectures. The normalized fidelity metrics divide by a base-to-counterfactual input-effect scale. Table 11 reports the release-minus-retention difference in that denominator for each architecture on the primary `two_concepts` task. The probability-space quantity is $E_{\text{prob}}(0, 1)$ from Equation (4); the centered-logit quantity is its direct analogue after centering logits. Each estimate averages the four initialization replicates within renderer block before inference over 100 blocks. The wide variation across architectures is one reason the main text reports the unnormalized probability-error reduction alongside normalized fidelity.

Table 11: Release-minus-retention differences in the base-to-counterfactual input-effect denominator on `two_concepts`. Intervals are paired 95% t intervals over 100 renderer blocks.

Architecture	$\Delta E_{\text{prob}}(0, 1)$	95% CI	$\Delta E_{\text{clogit}}(0, 1)$	95% CI
<code>tiny_gmp</code>	+0.83752	[+0.83054, +0.84450]	+126.31	[+124.45, +128.18]
<code>tiny_gap</code>	+0.19010	[+0.18295, +0.19724]	+5.34	[+5.07, +5.62]
<code>plain2_w16_gmp</code>	+0.06108	[+0.05866, +0.06350]	+204.42	[+201.23, +207.60]
<code>plain2_w16_gap</code>	+0.74200	[+0.73086, +0.75315]	+216.93	[+208.71, +225.15]
<code>plain4_w16_gmp</code>	+0.00472	[+0.00371, +0.00573]	+181.74	[+168.61, +194.86]
<code>plain4_w16_gap</code>	+0.13850	[+0.12074, +0.15627]	+601.26	[+548.89, +653.63]
<code>plain2_w64_gmp</code>	+0.02535	[+0.02383, +0.02686]	+186.40	[+183.23, +189.56]
<code>plain2_w64_gap</code>	+0.58107	[+0.57304, +0.58910]	+181.49	[+175.74, +187.25]
<code>plain4_w64_gmp</code>	-0.00027	[-0.00053, -0.00002]	+85.05	[+77.20, +92.91]
<code>plain4_w64_gap</code>	+0.06385	[+0.04470, +0.08301]	+625.91	[+558.30, +693.51]
<code>res4_w16_gmp</code>	+0.00298	[+0.00229, +0.00367]	+132.71	[+124.93, +140.48]
<code>res4_w16_gap</code>	+0.12994	[+0.11077, +0.14911]	+553.74	[+516.93, +590.55]
<code>bn2_w16_gmp</code>	+0.08357	[+0.06440, +0.10274]	+122.19	[+116.66, +127.72]
<code>bn2_w16_gap</code>	+0.49681	[+0.45913, +0.53449]	+57.03	[+55.08, +58.98]
<code>bn4_w16_gmp</code>	+0.00290	[+0.00238, +0.00341]	+6.68	[+5.46, +7.91]
<code>bn4_w16_gap</code>	+0.02535	[+0.01827, +0.03243]	+8.26	[+7.50, +9.02]

B.3 Architecture-specific estimates

Table 12 reports every architecture mean and marginal 95% t interval for the two pre-specified evaluation metrics.

Table 12: `two_concepts` architecture means for B .

Architecture	B_{clogit}	95% CI	$B_{R_{\text{prob}}}$	95% CI
tiny_gmp	+0.1456	[+0.1352, +0.1560]	+0.8315	[+0.8122, +0.8508]
tiny_gap	+0.0849	[+0.0764, +0.0934]	+0.0944	[+0.0901, +0.0986]
plain2_w16_gmp	-0.0167	[-0.0227, -0.0106]	-0.0581	[-0.0757, -0.0404]
plain2_w16_gap	+0.0936	[+0.0734, +0.1139]	+0.3348	[+0.3207, +0.3488]
plain4_w16_gmp	-0.0168	[-0.0211, -0.0124]	-0.0561	[-0.0675, -0.0447]
plain4_w16_gap	+0.0277	[+0.0138, +0.0416]	+0.1160	[+0.0992, +0.1328]
plain2_w64_gmp	-0.0072	[-0.0113, -0.0031]	-0.0021	[-0.0152, +0.0111]
plain2_w64_gap	+0.2213	[+0.2022, +0.2404]	+0.3247	[+0.3045, +0.3449]
plain4_w64_gmp	-0.0049	[-0.0068, -0.0029]	-0.0238	[-0.0282, -0.0194]
plain4_w64_gap	+0.0511	[+0.0408, +0.0614]	+0.0013	[-0.0150, +0.0176]
res4_w16_gmp	-0.0175	[-0.0216, -0.0133]	-0.0448	[-0.0554, -0.0342]
res4_w16_gap	+0.0444	[+0.0276, +0.0611]	+0.0974	[+0.0756, +0.1192]
bn2_w16_gmp	+0.0165	[+0.0024, +0.0306]	+0.0127	[-0.0128, +0.0382]
bn2_w16_gap	-0.1382	[-0.1960, -0.0804]	-0.0390	[-0.0706, -0.0074]
bn4_w16_gmp	+0.0132	[+0.0063, +0.0201]	-0.0023	[-0.0163, +0.0116]
bn4_w16_gap	+0.1445	[+0.1177, +0.1713]	+0.0757	[+0.0532, +0.0983]

B.4 Pre-specified architecture-factor contrasts

Table 13 gives all pre-specified factor contrasts on the primary `two_concepts` task. Holm adjustment is applied within metric, task, and factor family exactly as specified in the architecture protocol. The signs are alternative architecture minus reference architecture as named in each row.

Table 13: Architecture-factor contrasts.

Family	Contrast	ΔB_{clogit}	Holm p	$\Delta B_{R_{\text{prob}}}$	Holm p
pooling	tiny: GAP - GMP	-0.0607	7.25×10^{-14}	-0.7372	4.41×10^{-88}
pooling	plain2_w16: GAP - GMP	+0.1103	3.42×10^{-17}	+0.3929	3.70×10^{-56}
pooling	plain4_w16: GAP - GMP	+0.0445	7.53×10^{-8}	+0.1721	4.57×10^{-28}
pooling	plain2_w64: GAP - GMP	+0.2285	5.03×10^{-40}	+0.3268	2.56×10^{-46}
pooling	plain4_w64: GAP - GMP	+0.0560	1.60×10^{-17}	+0.0251	0.0073
pooling	res4_w16: GAP - GMP	+0.0618	2.17×10^{-10}	+0.1422	3.48×10^{-19}
pooling	bn2_w16: GAP - GMP	-0.1547	1.17×10^{-6}	-0.0517	0.0146
pooling	bn4_w16: GAP - GMP	+0.1313	4.09×10^{-15}	+0.0780	1.34×10^{-7}
depth	GMP: plain2 - tiny	-0.1623	6.62×10^{-45}	-0.8896	6.48×10^{-82}
depth	GAP: plain2 - tiny	+0.0087	0.6905	+0.2404	4.30×10^{-57}
depth	GMP: plain4 - plain2	-0.0001	0.9845	+0.0019	0.8552
depth	GAP: plain4 - plain2	-0.0659	6.12×10^{-7}	-0.2188	1.41×10^{-36}
width	depth2 GMP: w64 - w16	+0.0095	0.0095	+0.0560	1.40×10^{-6}
width	depth2 GAP: w64 - w16	+0.1277	1.51×10^{-17}	-0.0100	0.3637
width	depth4 GMP: w64 - w16	+0.0119	2.14×10^{-6}	+0.0323	1.82×10^{-6}
width	depth4 GAP: w64 - w16	+0.0234	0.0095	-0.1147	8.27×10^{-16}
residual	GMP: residual4 - plain4	-0.0007	0.7775	+0.0113	0.1961
residual	GAP: residual4 - plain4	+0.0166	0.1943	-0.0186	0.1961
batchnorm	depth2 GMP: BN - noBN	+0.0332	1.09×10^{-5}	+0.0708	3.31×10^{-5}
batchnorm	depth2 GAP: BN - noBN	-0.2318	2.67×10^{-11}	-0.3738	1.92×10^{-37}
batchnorm	depth4 GMP: BN - noBN	+0.0299	4.90×10^{-11}	+0.0538	2.61×10^{-7}
batchnorm	depth4 GAP: BN - noBN	+0.1168	2.67×10^{-11}	-0.0403	0.0036

The residual-versus-plain contrasts give the clearest null results: all four metric/pooling combinations have intervals spanning zero after the pre-specified Holm correction. Pooling, depth, width, and BatchNorm show multiple nonzero contrasts, but directions vary by backbone and metric. The data therefore support architecture dependence without supporting a single architecture-independent component effect.

B.5 Secondary `single_shape` architecture results

The secondary task also shows substantial variation across architectures, but its sign pattern differs in several settings. Table 14 is included to make that task dependence visible rather than pooling the two tasks.

For the architecture experiment, the maximum full-patch and no-op logit identity errors are zero. Initialization-dispersion summaries, denominator contrasts, random-channel controls, replacement-energy controls, and block-level values are retained in the versioned analysis tables.

Table 14: `single_shape` mean B by architecture.

Architecture	B_{clogit}	$B_{R_{\text{prob}}}$
tiny_gmp	+0.0569	+0.3459
tiny_gap	-0.0135	+0.2644
plain2_w16_gmp	+0.0525	-0.0203
plain2_w16_gap	+0.0449	+0.2026
plain4_w16_gmp	+0.0143	+0.0115
plain4_w16_gap	+0.1551	+0.1475
plain2_w64_gmp	+0.0321	-0.0463
plain2_w64_gap	+0.0432	+0.1525
plain4_w64_gmp	+0.0021	+0.0015
plain4_w64_gap	+0.0808	+0.0923
res4_w16_gmp	+0.0139	+0.0194
res4_w16_gap	+0.1539	+0.1241
bn2_w16_gmp	+0.0668	+0.1155
bn2_w16_gap	-0.2701	-0.0340
bn4_w16_gmp	+0.0028	+0.0102
bn4_w16_gap	+0.3773	+0.0383

C Pre-specified intervention-size and schedule/task sensitivity analyses

This appendix reports the full numerical results for the pre-specified intervention-size experiment and the subsequent schedule/task sensitivity experiment. The intervention-size protocol was specified after the earlier analyses but before running blocks 4000–4019, which are disjoint from the earlier data. It was not externally preregistered. The schedule/task sensitivity protocol was separately specified before running blocks 5000–5049 and remains secondary to the intervention-size test. Blocks are independent experimental replicates; conditions within a block are paired through shared generated data and relevant initialization components. The main text reports the primary intervention-size result and uses the schedule/task analysis as a secondary sensitivity analysis. This appendix preserves the complete channel-selection and schedule breakdowns, secondary controls, and reimplementations details. The complete tables are retained under `analysis/budget_confirmation_001/` and `analysis/exhaustive_robustness_001/`.

C.1 Pre-specified selected-channel patching results

Each entry is the within-block difference in patching effect for selected channels at epoch 200 (annealed minus constant regularization), averaged over 20 independent blocks. Only TinyCNN with channels ranked by activation difference supplies the single primary pre-specified test; the remaining rows are pre-specified secondary analyses.

Table 15: Annealed-minus-constant differences in patching effect for selected channels.

Architecture	Channel selection	k	Mean ΔF	95% t interval
TinyCNN	activation difference	1	-0.3459	[-0.4033, -0.2885]
TinyCNN	activation difference	2	-0.3317	[-0.4082, -0.2552]
TinyCNN	activation difference	4	-0.0221	[-0.0424, -0.0019]
TinyCNN	activation difference	8	+0.0065	[+0.0037, +0.0093]
TinyCNN	AUROC	1	-0.3040	[-0.3662, -0.2418]
TinyCNN	AUROC	2	-0.3237	[-0.3988, -0.2485]
TinyCNN	AUROC	4	-0.0332	[-0.0696, +0.0032]
TinyCNN	AUROC	8	+0.0058	[+0.0026, +0.0091]
TinyCNN	validation patching	1	-0.2856	[-0.3191, -0.2520]
TinyCNN	validation patching	2	-0.1733	[-0.2068, -0.1399]
TinyCNN	validation patching	4	-0.0052	[-0.0202, +0.0098]
TinyCNN	validation patching	8	+0.0073	[+0.0047, +0.0098]
TwoLayerCNN	activation difference	1	-0.0117	[-0.0202, -0.0032]
TwoLayerCNN	activation difference	2	-0.0378	[-0.0493, -0.0263]
TwoLayerCNN	activation difference	4	-0.1313	[-0.1645, -0.0980]
TwoLayerCNN	activation difference	8	-0.0020	[-0.0493, +0.0453]
TwoLayerCNN	AUROC	1	-0.0141	[-0.0224, -0.0058]
TwoLayerCNN	AUROC	2	-0.0438	[-0.0565, -0.0311]
TwoLayerCNN	AUROC	4	-0.1319	[-0.1614, -0.1025]
TwoLayerCNN	AUROC	8	+0.0104	[-0.0402, +0.0611]
TwoLayerCNN	validation patching	1	-0.0256	[-0.0380, -0.0131]
TwoLayerCNN	validation patching	2	-0.0760	[-0.1048, -0.0473]
TwoLayerCNN	validation patching	4	-0.0883	[-0.1401, -0.0365]
TwoLayerCNN	validation patching	8	+0.0100	[-0.0015, +0.0214]

C.2 Pre-specified intervention-size contrast

For every architecture/channel-selection pair,

$$B = \frac{\Delta F(4) + \Delta F(8)}{2} - \frac{\Delta F(1) + \Delta F(2)}{2}. \quad (17)$$

Positive B indicates that the annealed-minus-constant difference is larger on average at $k = 4, 8$ than at $k = 1, 2$; it does not imply monotonicity in k and does not require a positive annealing effect at either larger value. The first TinyCNN row, with channels ranked by activation difference, is the single pre-specified primary test. Its 20 block-level B values are all positive (range 0.1452–0.5986). All other rows are secondary and do not alter the primary conclusion.

Table 16: Values of the pre-specified contrast B . The first row alone is the primary pre-specified test.

Architecture	Channel selection	Mean B	95% CI	two-sided p
TinyCNN	activation difference	+0.33098	[0.27368, 0.38827]	2.28×10^{-10}
TinyCNN	AUROC	+0.30014	[0.24866, 0.35162]	1.95×10^{-10}
TinyCNN	validation patching	+0.23050	[0.20442, 0.25658]	1.31×10^{-13}
TwoLayerCNN	activation difference	-0.04189	[-0.07010, -0.01367]	0.00580
TwoLayerCNN	AUROC	-0.03179	[-0.05830, -0.00529]	0.02126
TwoLayerCNN	validation patching	+0.01162	[-0.00949, 0.03273]	0.26368

C.3 Patched-run accuracy on counterfactual targets

Patched-run accuracy on the counterfactual target was retained alongside the normalized patching effect. For the primary TinyCNN setting, with channels ranked by activation difference, annealed minus constant regularization is -0.2009 at $k = 1$, -0.3626 at $k = 2$, -0.0241 at $k = 4$, and $+0.0140$ at $k = 8$. The corresponding 95% intervals are $[-0.2551, -0.1466]$, $[-0.4400, -0.2852]$, $[-0.0524, 0.0042]$, and $[0.0086, 0.0193]$.

AUROC and validation-patching channel-selection rules show the same broad change between smaller and larger channel sets. These are secondary outcomes and were not used to define the primary success criterion.

C.4 Energy-matched control

The energy-matched control uses validation replacement energy to select eight alternative channel sets of the same size whose replacement energy is closest to that of the selected set, using validation data only. Here F denotes the original probability-fidelity score in Equation (5). For a fixed model, concept, and intervention size k , let

$$F_{\text{random}}(k) = \frac{1}{8} \sum_{r=1}^8 F(S_{\text{random}}^{(r)}, k), \quad F_{\text{energy}}(k) = \frac{1}{8} \sum_{r=1}^8 F(S_{\text{energy}}^{(r)}, k), \quad (18)$$

where the first average uses eight same-size random channel sets and the second uses the eight validation-energy-matched sets. The relative control quantities are

$$U_{\text{random}}(k) = F_{\text{selected}}(k) - F_{\text{random}}(k), \quad U_{\text{energy}}(k) = F_{\text{selected}}(k) - F_{\text{energy}}(k). \quad (19)$$

Thus both quantities compare exactly the same selected-channel patching score with an average over eight alternative channel sets; only the control-set construction changes. For TinyCNN with channels ranked by activation difference, the annealed-minus-constant difference in U_{energy} is -0.2980 at $k = 1$, -0.1887 at $k = 2$, $+0.1310$ at $k = 4$, and $+0.0696$ at $k = 8$. Matching quality is not uniform: single-channel alternatives can be poorly matched, whereas $k = 4$ and $k = 8$ are generally much tighter. This analysis shows that the difference relative to the energy-matched baseline also changes with intervention size, but the primary analysis remains the patching effect for selected channels.

C.5 Schedule sensitivity when channels are ranked by activation difference

The main text uses the schedule/task experiment only as a secondary sensitivity analysis. The table below gives all five schedule-versus-constant-regularization comparisons for each task/architecture setting. Holm adjustment is performed within each task/architecture/channel-selection family, exactly as pre-specified.

Table 17: Schedule-sensitivity values of B for every non-reference regularization schedule when channels are ranked by activation difference.

Task	Architecture	Schedule vs. constant $\lambda = 1$	Mean B	95% CI	Holm p
single shape	TinyCNN	initialization only	+0.06804	[0.05343, 0.08264]	6.91×10^{-12}
single shape	TinyCNN	constant $\lambda = 0.1$	+0.03871	[0.02525, 0.05218]	5.13×10^{-7}
single shape	TinyCNN	early annealing	+0.06855	[0.05228, 0.08482]	1.11×10^{-10}
single shape	TinyCNN	default annealing	+0.07189	[0.05444, 0.08934]	1.43×10^{-10}
single shape	TinyCNN	late annealing	+0.07955	[0.06404, 0.09505]	3.61×10^{-13}
single shape	TwoLayerCNN	initialization only	+0.00762	[-0.00793, 0.02317]	0.9885
single shape	TwoLayerCNN	constant $\lambda = 0.1$	+0.00031	[-0.01053, 0.01116]	1.0000
single shape	TwoLayerCNN	early annealing	+0.00003	[-0.01500, 0.01506]	1.0000
single shape	TwoLayerCNN	default annealing	-0.01018	[-0.02290, 0.00254]	0.4569
single shape	TwoLayerCNN	late annealing	-0.01237	[-0.02194, -0.00279]	0.0621
two concepts	TinyCNN	initialization only	+0.34557	[0.31797, 0.37317]	5.61×10^{-29}
two concepts	TinyCNN	constant $\lambda = 0.1$	+0.16626	[0.14002, 0.19251]	3.71×10^{-17}
two concepts	TinyCNN	early annealing	+0.32245	[0.28415, 0.36074]	1.26×10^{-21}
two concepts	TinyCNN	default annealing	+0.28619	[0.25389, 0.31850]	1.95×10^{-22}
two concepts	TinyCNN	late annealing	+0.19342	[0.16432, 0.22253]	1.20×10^{-17}
two concepts	TwoLayerCNN	initialization only	-0.00583	[-0.03214, 0.02049]	0.6583
two concepts	TwoLayerCNN	constant $\lambda = 0.1$	-0.01413	[-0.03880, 0.01054]	0.5104
two concepts	TwoLayerCNN	early annealing	-0.04199	[-0.06889, -0.01509]	0.0116
two concepts	TwoLayerCNN	default annealing	-0.04156	[-0.06983, -0.01329]	0.0144
two concepts	TwoLayerCNN	late annealing	-0.04523	[-0.06854, -0.02192]	0.00147

C.6 Schedule/task sensitivity endpoints at epoch 200

The comparison below complements the activation-patching results with test accuracy and first-layer filter-template similarity. Values are means over 50 new replicates.

Table 18: Schedule/task sensitivity test accuracy and first-layer filter-template similarity for default annealing versus constant regularization.

Task	Architecture	Schedule	Test accuracy	Template similarity
single shape	TinyCNN	constant $\lambda = 1$	99.98%	0.958
single shape	TinyCNN	default annealing	99.99%	0.682
single shape	TwoLayerCNN	constant $\lambda = 1$	99.98%	1.000
single shape	TwoLayerCNN	default annealing	99.99%	0.942
two concepts	TinyCNN	constant $\lambda = 1$	98.05%	0.944
two concepts	TinyCNN	default annealing	99.45%	0.574
two concepts	TwoLayerCNN	constant $\lambda = 1$	99.11%	0.999
two concepts	TwoLayerCNN	default annealing	99.26%	0.896

C.7 Independent reimplementations of checkpoint evaluation

The pre-specified reimplementations covered 32 earlier checkpoint evaluations. All 32 completed after correcting an error in the independent implementation that initially ranked post-second-layer rather than first-layer activations in TwoLayerCNN. The correction did not alter the pre-specified protocol. All recomputed values then fell within the stated tolerances.

Table 19: Discrepancies between the original and independently reimplemented checkpoint evaluations. Full-channel and no-op probability errors were both zero.

Metric	Maximum absolute discrepancy	Tolerance
ordinary accuracy	0	10^{-12}
filter-template similarity	2.22×10^{-16}	10^{-6}
concept AUROC	0	10^{-9}
localization IoU	0	10^{-9}
patching effect for selected channels	2.39×10^{-7}	2×10^{-6}
patching effect for random channels	1.14×10^{-7}	2×10^{-6}
U_{random}	2.19×10^{-7}	2×10^{-6}
patched-run accuracy	0	10^{-12}

D Earlier analyses motivating the pre-specified contrast

These earlier analyses precede the pre-specified intervention-size experiment reported in Section C. They are retained to show why the primary analysis used the patching effect for selected channels rather than the selected-minus-random comparison.

D.1 Early 40-epoch comparisons

The initial 40-epoch analysis compared constant template regularization (`template_retention_1`) with template initialization only (`template_init`) at 40 epochs. Four tests of $U_{\text{random}} = F_{\text{selected}} - F_{\text{random}}$ were specified before the outcomes were inspected and were Holm-adjusted as one family; Equations (18) and (19) gives the later explicit definition of the same selected-minus-random quantity, including the eight-control average. Filter-template similarity and accuracy are also reported descriptively.

Table 20: Early four-comparison analysis. The positive marginal interval in the first row does not survive Holm adjustment.

Task	Architecture	Δ template similarity	Δ acc. (pp)	ΔU [95% CI]	Holm p
single shape	TinyCNN	+0.268	-0.23	+0.0345 [0.0018, 0.0672]	0.1633
single shape	TwoLayerCNN	+0.172	-0.04	-0.0092 [-0.0513, 0.0329]	1.0000
two concepts	TinyCNN	+0.289	-6.13	+0.0252 [-0.0306, 0.0811]	0.9995
two concepts	TwoLayerCNN	+0.198	-0.39	-0.0074 [-0.0557, 0.0408]	1.0000

D.2 Long-horizon epoch-200 condition means

The table below gives the earlier epoch-200 quantities most relevant to the constant-versus-annealed regularization comparison. Means are over ten replicates. U_{random} is the selected-channel probability fidelity minus the mean fidelity over eight same-size random channel sets, as defined in Equations (18) and (19). The full five-condition tables and checkpoint trajectories remain in the analysis artifact.

Table 21: Earlier epoch-200 means for constant and annealed template regularization.

Task	Architecture	Condition	Accuracy (%)	Template similarity	U_{random}
single shape	TinyCNN	constant $\lambda = 1$	99.961	0.959	0.358
single shape	TinyCNN	annealed	100.000	0.690	0.335
single shape	TwoLayerCNN	constant $\lambda = 1$	99.922	1.000	0.067
single shape	TwoLayerCNN	annealed	99.961	0.942	0.060
two concepts	TinyCNN	constant $\lambda = 1$	98.008	0.947	0.657
two concepts	TinyCNN	annealed	99.219	0.580	0.718
two concepts	TwoLayerCNN	constant $\lambda = 1$	99.141	0.999	0.061
two concepts	TwoLayerCNN	annealed	99.531	0.907	0.003

On `two_concepts`/TinyCNN, patched-run accuracy on the counterfactual target is 93.61% under constant regularization and 90.64% under annealed regularization even though U_{random} is higher under annealing. This discrepancy motivated separate reporting of the selected-channel and control-channel patching effects.

D.3 Post-hoc decomposition of the original selected-minus-random comparison

This post-hoc analysis reused the 80 earlier epoch-200 checkpoints from the annealed- and constant-regularization conditions and evaluated three validation-only channel-selection rules at $k \in \{1, 2, 4, 8\}$. The TinyCNN results are reproduced below. Each entry is annealed minus constant regularization; intervals are marginal paired 95% t intervals over ten replicates. This analysis is post-hoc and is not used as the primary pre-specified test.

Table 22: Post-hoc `two_concepts`/TinyCNN decomposition.

Channel selection	k	ΔF_{sel}	ΔF_{rand}	ΔU	Δ CF acc.
activation difference	1	-0.375	-0.040	-0.335	-0.255
activation difference	2	-0.163	-0.062	-0.101	-0.185
activation difference	4	-0.022	-0.083	+0.061	-0.030
activation difference	8	+0.006	-0.033	+0.040	+0.013
AUROC	1	-0.332	-0.040	-0.292	-0.230
AUROC	2	-0.188	-0.062	-0.125	-0.218
AUROC	4	-0.022	-0.083	+0.061	-0.031
AUROC	8	+0.007	-0.033	+0.040	+0.015
validation patching	1	-0.277	-0.040	-0.237	-0.194
validation patching	2	-0.122	-0.062	-0.060	-0.147
validation patching	4	-0.015	-0.083	+0.067	-0.021
validation patching	8	+0.005	-0.033	+0.038	+0.013

When channels are ranked by activation difference, $\Delta F_{sel}(4) = -0.022$ with 95% interval $[-0.054, 0.010]$, while $\Delta F_{rand}(4) = -0.083$ with interval $[-0.097, -0.068]$. The positive $\Delta U(4) = +0.061$ is therefore partly due to the change in the random-channel baseline. At $k = 8$, the patching effect for selected channels itself becomes slightly positive: $+0.006$ $[0.004, 0.009]$. AUROC and validation-patching channel selection show the same broad pattern.

For `two_concepts`/TwoLayerCNN, the patching effect for selected channels at $k = 4$ is negative under all three channel-selection rules (approximately -0.125, -0.121, and -0.087 for activation difference, AUROC, and validation patching, respectively). Under validation-patching selection the score becomes positive at $k = 8$ ($+0.0215$ with marginal interval $[0.0049, 0.0381]$). These outcomes motivated the later pre-specified contrast between smaller ($k = 1, 2$) and larger ($k = 4, 8$) channel sets rather than a hypothesis about the sign at any single k .

D.4 Validation-energy-matched control

A later post-hoc control chose alternative channel sets of the same size whose validation replacement energy was closest to that of the selected set. All 80 checkpoint evaluations were completed. Matching quality varies substantially with intervention size: single-channel sets can have large relative mismatch, while $k = 4$ and especially $k = 8$ are generally tightly matched in the `two_concepts`/TinyCNN setting. Changing the baseline does not alter the annealed-minus-constant effect for the selected channels, which is the quantity used to motivate the pre-specified intervention-size experiment. The comparison with the energy-matched baseline is therefore reported as a secondary analysis.

D.5 Transition to the pre-specified intervention-size experiment

The earlier analyses showed that constant template regularization preserves high filter-template similarity, that prediction can improve after the regularizer is annealed to zero, and that the annealed-minus-constant patching comparison differed across the tested intervention sizes. Because the original sign change in U combined changes in both the selected-channel and baseline patching effects, the intervention-size protocol was specified around the patching effect for selected channels before the new replicates were generated. Section C reports that pre-specified test and the subsequent secondary sensitivity analysis over 50 new blocks per task/architecture setting (1,200 trained models in total).

E Complete filter-matching table

The revised main text uses the one-to-one matching analysis only as a compact structural robustness check. This appendix reports the complete epoch-200 numerical table; the archived research artifact also retains the matching trajectories, full similarity matrices, and matched-filter gallery from the earlier manuscript version.

All values below are epoch-200 means $\pm$ sample standard deviation across ten replicates; conditions are paired within replicate. The nearest-template and one-to-one scores use signed cosine similarity. Coverage counts distinct nearest-template entries; because the bank has rank 10, it is neither a count of linearly independent directions nor a count of independent semantic concepts. The analysis uses only saved checkpoints from the earlier analyses. These descriptive comparisons are exploratory.

Table 23: Epoch-200 filter-template matching results.

Task	Architecture	Condition	Nearest-template	One-to-one	Coverage
single shape	TinyCNN	random	0.300 ± 0.023	0.267 ± 0.033	9.700 ± 1.703
single shape	TinyCNN	unit-norm random	0.288 ± 0.022	0.256 ± 0.034	9.700 ± 1.703
single shape	TinyCNN	initialization only	0.610 ± 0.021	0.605 ± 0.021	13.300 ± 1.337
single shape	TinyCNN	annealed regularization	0.690 ± 0.017	0.686 ± 0.017	14.200 ± 1.033
single shape	TinyCNN	constant $\lambda = 1$ regularization	0.959 ± 0.003	0.959 ± 0.003	16.000 ± 0.000
single shape	TwoLayerCNN	random	0.261 ± 0.037	0.230 ± 0.039	9.200 ± 1.317
single shape	TwoLayerCNN	unit-norm random	0.228 ± 0.033	0.195 ± 0.039	8.300 ± 1.889
single shape	TwoLayerCNN	initialization only	0.799 ± 0.015	0.799 ± 0.015	15.700 ± 0.483
single shape	TwoLayerCNN	annealed regularization	0.942 ± 0.006	0.942 ± 0.006	16.000 ± 0.000
single shape	TwoLayerCNN	constant $\lambda = 1$ regularization	1.000 ± 0.000	1.000 ± 0.000	16.000 ± 0.000
two concepts	TinyCNN	random	0.232 ± 0.034	0.199 ± 0.035	9.100 ± 1.524
two concepts	TinyCNN	unit-norm random	0.222 ± 0.031	0.191 ± 0.037	9.200 ± 1.398
two concepts	TinyCNN	initialization only	0.507 ± 0.022	0.496 ± 0.022	12.000 ± 1.054
two concepts	TinyCNN	annealed regularization	0.580 ± 0.011	0.574 ± 0.010	12.900 ± 0.994
two concepts	TinyCNN	constant $\lambda = 1$ regularization	0.947 ± 0.004	0.947 ± 0.004	16.000 ± 0.000
two concepts	TwoLayerCNN	random	0.255 ± 0.027	0.231 ± 0.024	9.600 ± 1.075
two concepts	TwoLayerCNN	unit-norm random	0.243 ± 0.028	0.218 ± 0.027	9.000 ± 1.054
two concepts	TwoLayerCNN	initialization only	0.746 ± 0.017	0.746 ± 0.017	15.400 ± 0.699
two concepts	TwoLayerCNN	annealed regularization	0.907 ± 0.012	0.907 ± 0.012	16.000 ± 0.000
two concepts	TwoLayerCNN	constant $\lambda = 1$ regularization	0.999 ± 0.000	0.999 ± 0.000	16.000 ± 0.000